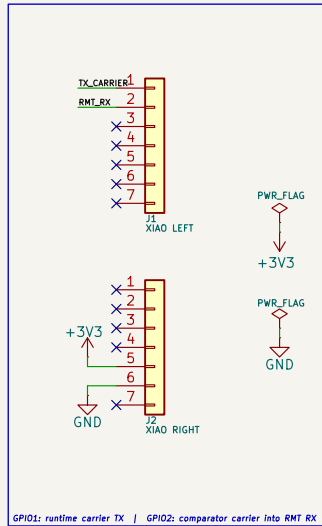
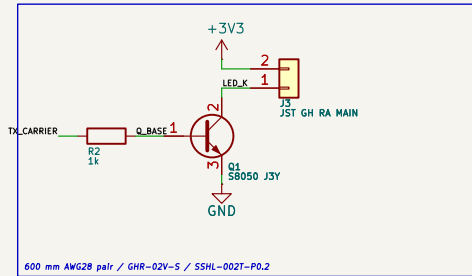


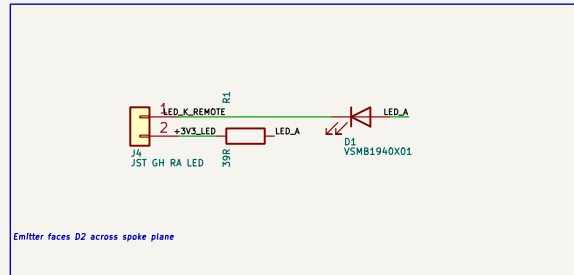
1 ESP32-S3 HOST / RMT



2 DRIVER / LOCKING HARNESS



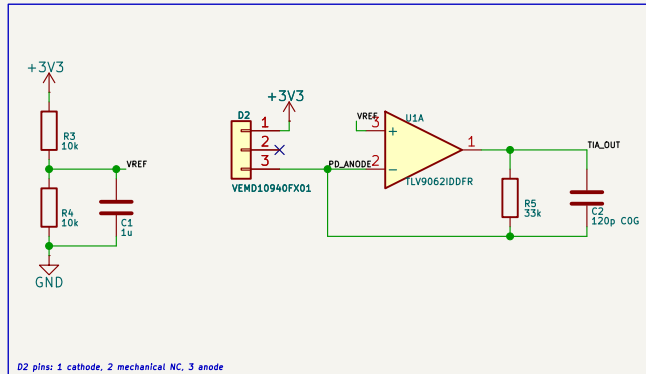
3 REMOTE EMITTER PCB



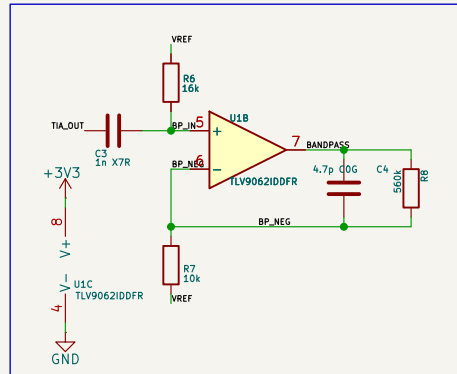
7 DEBUG / CORRELATION TEST POINTS



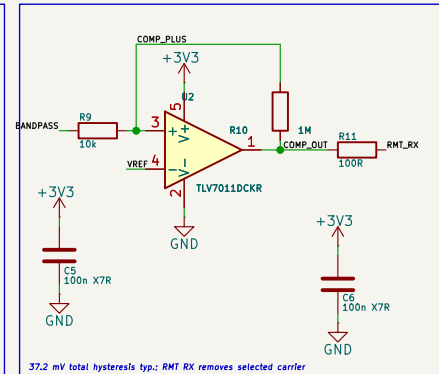
4 VREF / PHOTODIODE / TIA



5 10 kHz–60 kHz ACTIVE BAND-PASS



6 SCHMITT DECISION / RMT INPUT



SYSTEM NOTES
- RMT TX carrier is runtime-selectable from 25 kHz to 50 kHz.
- The discrete AFE rejects DC sunlight; the Schmitt stage restores logic.
- RMT RX carrier removal yields blockage intervals for the adaptive LUT.
- $\pm 3\text{V}_A$ are joined only by the defined 600 mm harness model.
- TP1-TP8 correlate physical nodes with transient traces.

BicycleOBU

Sheet: /

File: ir_spoke_link.kicad_sch

Title: IR Spoke Sensor – Discrete Optical Link

Size: A4

Date:

Rev: R4

KiCad E.D.A. 10.0.5

Id: 1/1